

Project Design Phase – II

Technology Stack (Architecture & Stack)

Date: 23 June 2026

Project Name: Metro Ticket Generating System in ServiceNow

Technical Architecture

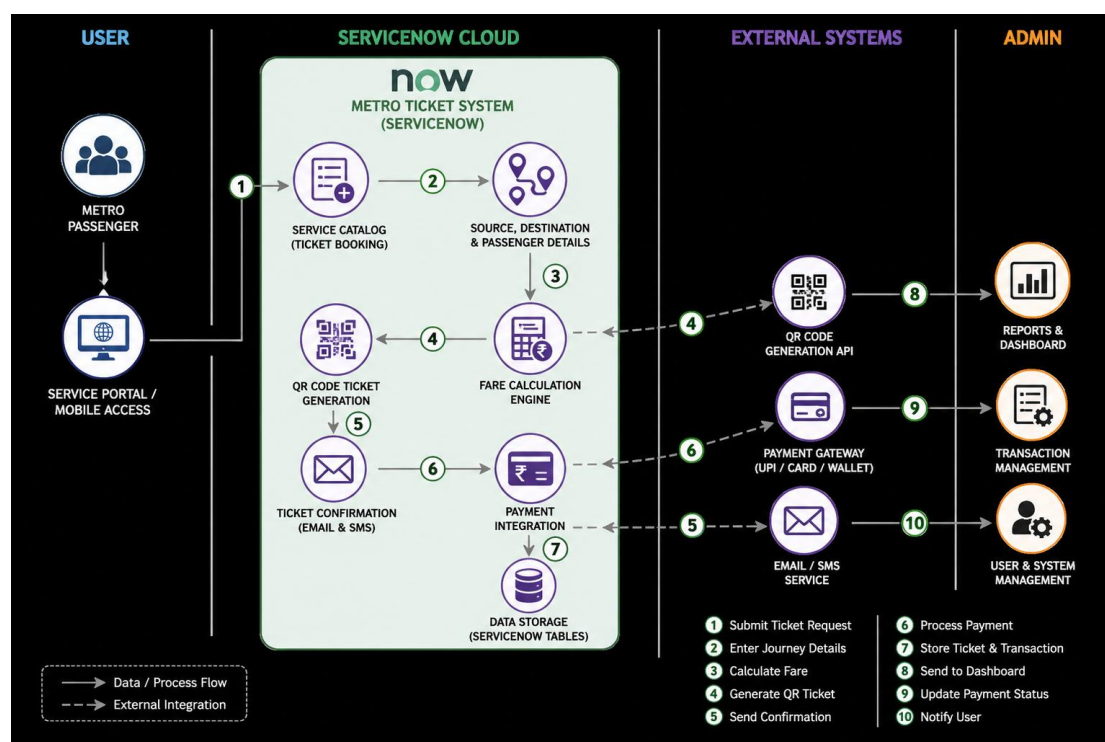


Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Passenger interacts with application through Service Portal	ServiceNow Service Portal, UI Builder
2	Application Logic-1	Metro ticket booking process	Service Catalog
3	Application Logic-2	Fare calculation and validations	Business Rules, Client Scripts
4	Application Logic-3	Ticket generation workflow	Flow Designer
5	Database	Ticket, Passenger and Station information	ServiceNow Tables
6	Cloud Database	Cloud-hosted application data	ServiceNow Cloud Platform

7	File Storage	QR code and ticket information storage	ServiceNow Attachments/Table Storage
8	External API-1	QR Code Generation	QR Code API / Scripted REST API
9	External API-2	Digital Payment Integration	UPI / Payment Gateway API
10	Machine Learning Model	Not Applicable for current scope	N/A
11	Infrastructure (Server / Cloud)	Application deployment and hosting	ServiceNow SaaS Cloud Platform

Table-2 : Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Web technologies used for portal customization	HTML, CSS, JavaScript
2	Security Implementations	Authentication, authorization and data protection	ServiceNow ACLs, Roles, HTTPS, RBAC
3	Scalable Architecture	Supports increasing passenger requests and transactions	ServiceNow Multi-Tenant Cloud Architecture
4	Availability	Application available anytime for ticket booking	ServiceNow High Availability Cloud Infrastructure
5	Performance	Fast ticket booking, fare calculation and QR generation	ServiceNow Platform Optimization, Indexed Tables

Infrastructure Details

Cloud Environment

- ServiceNow SaaS Platform
- ServiceNow Database
- Service Portal
- Flow Designer
- Business Rules
- Notification Engine
- QR Code Integration API

External Interfaces

- QR Code Generation API
- UPI Payment Gateway API
- Email Notification Service

Data Storage Components

- Metro Ticket Table
- Metro Station Table
- Fare Matrix Table
- Passenger Records

Application Logic Components

- Ticket Booking Module
- Fare Calculation Engine
- Ticket Generation Module
- QR Code Generation Module
- Notification Module
- Reporting & Analytics Module